ASSESSING THE DETECTION PERFORMANCE ON ICY TARGETS ACQUIRED BY AN ORBITING RADAR SOUNDER

Elena Donini[1,2], Sanchari Thakur[2], Francesca Bovolo[1], Lorenzo Bruzzone[2]

Fondazione Bruno Kessler, Trento, Italy[1], University of Trento, Trento, Italy[2]

ABSTRACT

Radar sounders (RS) can be used to acquire data on ice sheets and provide direct evidence of the structures in the subsurface. Many acquisitions are available from airborne RS in the Antarctica and Greenland. However, airborne data are costly, have limited spatial coverage, and non homogeneous characteristics. To overcome these limitations, a potential satellite-mounted RS could provide uniform coverage and consistent data quality at the cost of lower resolution and higher path loss. In this paper, we assess the performance of a possible Earth-orbiting RS by simulating and analyzing its radargrams. The simulation approach reprocesses existing airborne RS to match the orbital RS characteristics. The simulated radargrams are analyzed to estimate the losses and understand the detection performance of icy targets using state-of-the-art data analysis techniques. The preliminary analysis of the simulated radargrams indicates that, under the simplified assumptions, an orbiting RS will be capable of imaging the investigated subsurface targets.

Index Terms— Radar Sounder, Cryosphere, Radargram Simulation, Data Analysis

1. INTRODUCTION

The dynamic changes in the cryosphere due to global warming require continuous monitoring of the structures and the processes occurring within the ice-sheets. This information can be extracted from the vertical profiles of the ice subsurface (radargrams) acquired by RS. Subsurface targets detected from radargrams are used to determine: (i) the surface and the basal interface topography, ii) the ice sheet thickness and the internal layering , (iii) the characteristics of ice shelves and the position of the grounding line, (iv) the basal boundary conditions and ice flow regime, and (v) the ice mass balance.

Over the last decades, several airborne RS acquired a large amount of data in the Earth's icy areas [1]. However, these data are collected seasonally, over specific regions and with different instruments. Thus, they lack uniform coverage in time and space, and consistent quality. Moreover, the airborne radargrams show artifacts due to the low flight altitude and the instability of the platform.

A satellite-mounted RS would address these limitations, providing continuous, homogeneous, and uniform coverage of the Earth cryosphere. STRATUS (SaTellite Radar sounder for eArth sUb-surface Sensing) [2], a possible Earth-orbiting RS, is designed to operate at a central frequency in the HF-VHF range with a bandwidth of 10 MHz, to have a penetration depth up to 5 km and a vertical resolution of about 10 m in ice. To study the performance of an orbiting RS, it is necessary to consider the effects of the cosmic microwave background (CMB), the path loss, relatively low transmit power, the ionospheric distortions, and the relatively poor resolution.

One way to investigate these limitations is to simulate the orbital radargrams. Recently, techniques such as finite difference time domain (FDTD) [3] and ray tracing have been used for simulation and performance prediction of planetary RS data. However, conventional electromagnetic techniques require a subjective modeling of the target, which is challenging in case of complex targets, such as layered ice sheets.

This paper presents a simulation approach that is then used for a preliminary analysis of the performance of an orbiting RS imaging relevant cryosphere targets. The proposed approach simulates the orbital radargrams by reprocessing available airborne data over the considered target areas. This approach avoids subjective modeling by extracting the target models directly from the real acquisitions. Then, we analyze the quality of the simulated radargrams to assess the performance of the orbiting RS, and we study the detectability of the subsurface targets with existing automatic techniques defined for airborne data. Here, we develop the analysis considering three relevant types of targets in the cryosphere, i.e., englacial layering, sub-glacial lakes, and ice in coastal areas. We conclude that the satellite data can be used to extract information on the considered subsurface targets with nearly comparable performance to the airborne data.

2. SIMULATION OF THE ORBITING RADAR SOUNDER DATA

This section presents the approach for simulating the radargrams of an Earth-orbiting RS. Figure 1 shows the procedure for the simulations. Table 1 reports the parameters of the considered orbiting and airborne RS. The 45 MHz central frequency of the orbiting RS is suitable for the scientific require-

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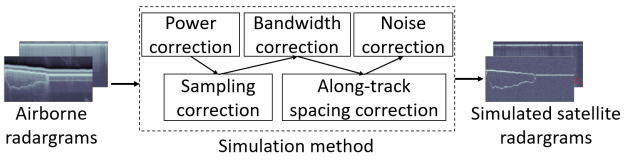


Fig. 1. Flowchart of the simulation and analysis method.

Table 1. Proposed orbiting and airborne RS parameters.

Parameter	Orbiting RS [2]	Airborne RS [1]
Central frequency	45 MHz	195 MHz
Bandwidth	10 MHz	30 MHz
Altitude	500 km	3.9 km
Transmitted Power	200 W	1050 W

ments of subsurface sounding in icy areas [2]. Although, currently the International Telecommunication Union (ITU) does not allocate this frequency for Earth observation, its allocation is included in the preliminary agenda of the 2023 World Radiocommunication Conference. Also, it provides a good balance between the penetration capability, the range resolution, the surface roughness, and the ionospheric effects [4].

The ionosphere interacts with the signal by introducing distortion in the radargrams that critically affects the phase and the range resolution. However, [4] shows that a signal with central frequency of 45 MHz and bandwidth of 10 MHz may propagate through the ionosphere under specific conditions and be able to image the cryosphere subsurface. These effects are not modeled in the proposed method, but they may be in the real radargrams, even after compensation.

Clutter is a critical quality issue for orbital radargrams, since the antenna footprint at the orbiting altitude is larger than at the airborne one. Hence, surface off-nadir echoes can mask weaker nadir subsurface reflections arriving at the same time. Currently, the proposed simulation approach does not model the presence of clutter due to both radar-scale surface roughness and off-nadir echoes. The airborne radargram is reprocessed by modelling and correcting the differences in the instrument parameters as follows.

Power correction: The normalized power (wrt surface) $P_s(z)$ depends on the target depth z, the platform altitude H, the subsurface reflectance Γ_{ss}, and the surface reflectivity R_s:

$$P_s(z) = \frac{\Gamma_{ss} H^2}{(H+z)^2 R_s} \tag{1}$$

z is calculated assuming the real ice permittivity equal to 3.15, and R_s is obtained using Fresnel equations. Γ_{ss} depends on the reflectivity R_{ss} at the subsurface dielectric interfaces and the power attenuation of the signal propagating through the ice medium. R_{ss} purely depends on the target properties, while the attenuation factor ($f_{c(a)}\alpha(z)$) depends on the instrument central frequency ($f_{c(a)}$) and a number of geophysical factors, and thus cannot be assumed constant. The subsurface reflectance in dB $\gamma_{ss(a)}$ is theoretically given by (2). We model $\gamma_{ss(a)}$ by fitting a polynomial function $p(z) = p_n z^n + ...p_1 z + p_0$ such that $p_0 = ln(R_{ss})$ and $p(z) - p_0 = f_{c(a)}\alpha(z)$. The subsurface reflectance in dB $\gamma_{ss(o)}$ of the orbital case is given by (3).

$$\gamma_{ss(a)} = ln(R_{ss}) + f_{c(a)}\alpha(z) \tag{2}$$

$$\gamma_{ss(o)} = \gamma_{ss(a)} + (p(z) - p_0)\left(\frac{f_{c(o)}}{f_{c(a)}} - 1\right) \tag{3}$$

For the results presented in this paper, a linear polynomial model was fitted to the $\gamma_{ss(a)}$. Note that, describing the attenuation rate by a polynomial function may introduce artifacts in the simulation, which should be considered while interpreting the simulated radargrams. The corrected power P_p is obtained using (4), where $\Gamma_{ss(o)}$ is the subsurface reflectance, P_t is the transmit power, G is the gain, λ is the wavelength, and H_o is the nominal orbit altitude of the orbiting RS:

$$P_p = \frac{P_t G^2 \lambda^2 \Gamma_{ss(o)}}{(4\pi)^3 (2(H_o + z))^2} \tag{4}$$

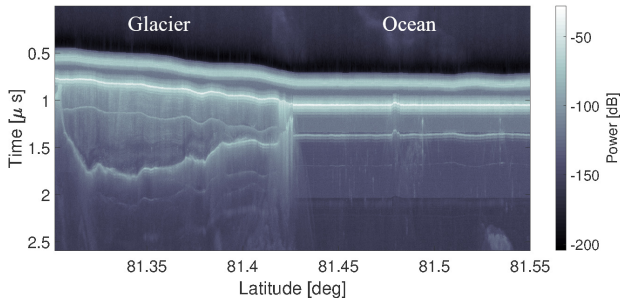
The considered orbiting RS, operating at a lower frequency than the airborne RS has a higher penetration depth due to lower power attenuation. However, at orbiting altitudes, there is a greater loss in power due to a longer propagation path.

Bandwidth correction: This requires low-pass filtering the power-corrected radargram, since the bandwidth of the airborne RS is larger than that of the orbiting RS. The bandwidth is inversely proportional to the range resolution, and thus the range resolution of the orbital radargrams will be lower.

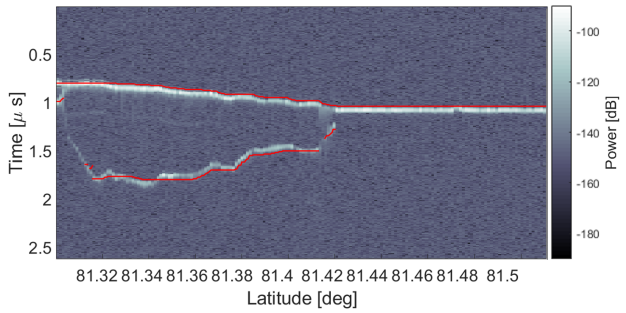
Sampling correction: The two-way time between successive samples depends on the sampling frequency. Accordingly, the bandwidth-corrected signal is resampled using the nearest neighbour interpolation, to obtain the temporal sampling of the orbiting RS.

Along-track spacing correction: The along-track support for the airborne radargram is obtained from the azimuth resolution of the airborne RS. This step requires uniformly resampling the rows of the radargram according to the lower azimuth resolution of the proposed orbiting RS.

Noise correction: Due to low flying altitudes of the airborne RS, the airborne radargrams are not affected by the isotropic cosmic microwave background (CMB) noise. However, this is present at orbiting altitudes and it average power can be estimated by $P_n = kBT_{eq,CMB}$, where $T_{eq,CMB}$ is the equivalent noise temperature of the CMB, B is the bandwidth of the orbiting RS receiver, and k is the Boltzmann constant. The noise component follows the Rayleigh distribution. We generate random Rayleigh noise P_n and add it to all the samples of the along-track spacing-corrected radargram. The result is the simulated orbital radargram. Note that in this approach, the information in simulated radargrams is



(a)



(b)

Fig. 2. (a) Airborne radargram 20170413_01_032-034 (b) Ice-bottom interface detected in red in the simulated radargram.

constrained by the information content of the airborne radargram.

3. ANALYSIS OF THE SIMULATED RADARGRAMS

In this section, we analyze the radargrams simulated from the data acquired by the MCoRDS radar sounder [1]. The data are range compressed, SAR focused, and processed using the minimum variance distortionless response beam former to reduce clutter. We simulated radargrams of 10 icy targets in coastal and inland areas. Here, we show the performance assessment of the orbital RS by presenting the preliminary results for three important targets– sub-glacial lakes, englacial layering, and ice interfaces in coastal areas.

The simulated radargrams are analyzed to study the effects of the orbiting RS parameters on the data quality and the performance in detecting the targets. The data quality is assessed by estimating the losses due to the propagation path in the air and in the medium. The losses in the medium include volume scattering losses due to heterogeneity caused by the layers and attenuation losses due to the dielectric profile. The detectability of the target is assessed by using state-of-the-art techniques already developed for airborne or satellite RS data. The automatic algorithms are selected by considering the state-of-the-art methods for the expected target.

Detection of the interfaces in coastal areas: Analyses of the roughness of Greenland and Antarctica at 45 MHz frequency show that the icy targets in the coastal areas are critical for orbital sounding [2] since the medium losses are found to be significantly higher than in other areas. Here, we analyze the subsurface losses of such a critical area using radargram 20170413_01_032-034 acquired in Greenland, Figure. 2 (a). The simulated orbital radargram (Fig. 2 (b)) shows floating ice in coastal areas, where the interface could either be ice/seawater or ice/bedrock.

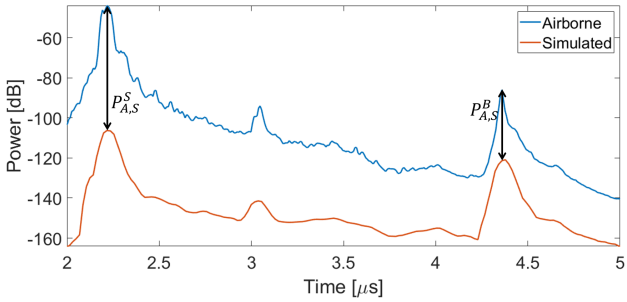


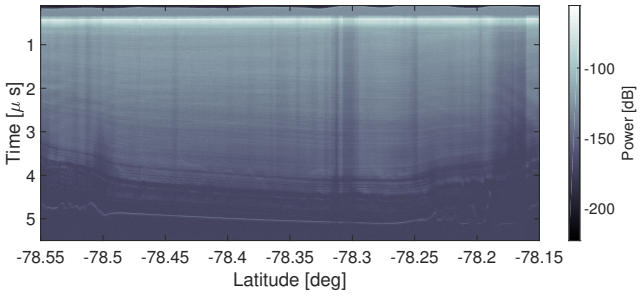
Fig. 3. A-scans of the airborne (20170413_01_032-034) and the simulated radargrams showing the path loss and the ice attenuation.

We use the simulated data to compare the power levels of the orbital and airborne RS. Figure 3 shows the absolute power profile of the airborne (orange) and the simulated (blue) radargrams. Due to the higher altitude of the orbital RS, the power of the surface $P^S_{A,S}$ in the simulated radargram is lower than that of the airborne. This mainly results from the altitude difference. The loss in power through the subsurface $P^B_{A,S}$, due to attenuation and volume scattering, is higher in the airborne case than the orbital one. Due to the longer wavelength of the orbital RS, and thus greater penetration capability, we obtain a compensation in the power received from the bottom. The power gain can be significantly improved by applying post processing, such as range compression and SAR processing [2]. The larger SAR gain is obtained from the larger antenna footprint of orbiting RS, and thus larger synthetic aperture length.

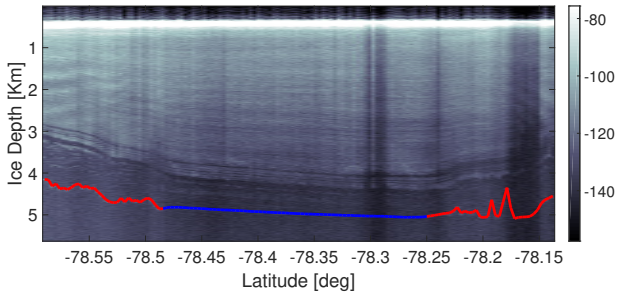
The interface detectability is critical for better understanding the melting of the ice and predicting the stability of the ice sheet. In the literature, this is done by visually analyzing the radargrams. Here, we propose to detect the interfaces with the algorithm presented in [5], which is meant for detecting layering in planetary data and is based on locally applying the Hidden Markov model and the Viterbi algorithm. Figure 2 (b) shows the detected interfaces in the simulated radargram.

Detection of subglacial lakes and englacial layers: Subglacial lakes lubricate the base of ice-sheets and speed-up the seaward flow of ice. Here, we present the analysis of the performance of the orbital RS, using the radargram acquired over lake Vostok. Figure 4 (a) shows the airborne radargram, while Figure 4 (b) shows the simulated radargram.

Figure 5 shows the loss profile (normalized wrt surface)



(a)



(b)

Fig. 4. (a) Airborne radargram 20131127_01_045-047 acquired in Antarctica, and (b) simulated radargram (subglacial lake in blue and bedrock in red).

of the ice-layering. We test the detectability of the layers by identifying prominent local maxima, marked by dots (red for the airborne and green for the simulated), on the profile. Due to the lower bandwidth of the orbital RS, hence the lower range resolution, a smaller number of layers are detected in the simulated radargram (108) than in the airborne (134).

Finally, we analyzed the detectability of the lake Vostok by applying the technique in [6]. The method extracts features related to the shape and statistical properties of the bedrock and uses them to train an SVM classifier. Figure 4 shows the results. Note that the flat reflector in blue is correctly classified as subglacial lake, while the red line is the non-lake interface. The overall accuracy is 94%, whereas in the airborne radargram is 96%.

4. CONCLUSION

In this paper, we present analyses of the performance of an orbiting RS for Earth observation in detecting subsurface cryosphere targets. An orbiting RS providing multitemporal, homogeneous, and uniform coverage would be extremely useful for the continuous monitoring of the cryosphere in the framework of climate change.

The preliminary assessment confirms that the considered orbiting RS would be able to image subsurface target, i.e., the interfaces in coastal and inland areas, since they are above the noise margin. Due to the longer wavelength of the orbiting RS, the average medium losses in the selected targets is $\sim$50 dB/km, compared to $\sim$76 dB/km in the airborne case. Thus, the bottom interface is still detectable and can be discriminated by means of existing data analysis algorithms into subglacial lake and bedrock. Further, the prominent layers are also detectable at the lower bandwidth, despite the lower range resolution.

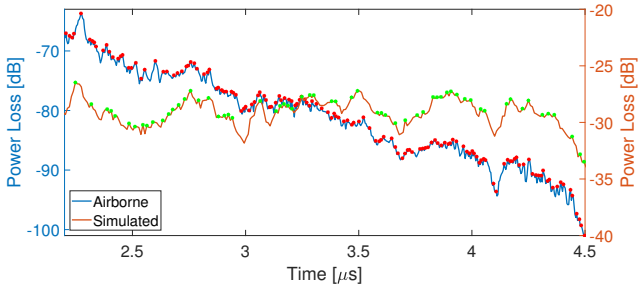


Fig. 5. Loss profile of the radargram 20131127_01_045-047 and the simulated one with the detected peaks highlighted with dots (possible layers). Different scales should be considered in the analysis of the behavior of the two curves.

As a future work, we plan to simulate a number of radargrams of critical icy targets, such as grounding zone, ice flows, and ice shelves, which are expected to have higher volume and surface scattering losses. Extending the experimental analysis can provide a reliable assessment of the feasibility in all areas and can support the design of the orbiting RS for Earth observation.

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